

Problem Statement 04: Ship Detection from Satellite Imagery

Background

Maritime Domain Awareness (MDA) requires continuous monitoring of vessel traffic across vast ocean areas. Satellite imagery provides wide-area coverage, but manual ship detection is time-consuming and error-prone.

Problem

Ship detection in optical satellite imagery is challenging due to varying lighting conditions, cloud cover, sea state, and the small size of vessels relative to image resolution. Automated detection systems must balance sensitivity (detecting all ships) with specificity (avoiding false positives from waves, clouds, and other artifacts).

Aim

Develop a computer vision model that detects ships in optical satellite images and predicts ship presence and ship count.

Output

1. Complete, reproducible training and inference source code (`train.py`, `predict.py`).
2. Exported model weights/checkpoints (`.pt` or `.onnx`).
3. Predictions CSV mapping `image_id` to ship presence classification and ship count.

Data

File Structure:

PS4_public/

├── train/

| ├── img_00001.jpg

| ├── img_00002.jpg

| └── ...

└── test| ├── img_05001.jpg

| └── img_05002.jpg
└── ...
└── train_labels.csv
└── sample_submission.csv

Data Description:

Images:

1. **Format:** JPG (RGB)
2. **Size:** 768 × 768 pixels
3. **Source:** Sentinel-2 satellite imagery
4. **Coverage:** Global shipping lanes, ports, coastal areas

train_labels.csv:

Column	Description	Type
image_id	Unique image identifier	int
image_path	Relative path to image	string
ship_present	Is there a ship? (0/1)	int
ship_count	Number of ships	int
ship_confidence	Detection confidence (0-1)	float

Note: Labels are only in train_labels.csv

Submission Format

File:

teamname_PS4.csv

1. image_id,ship_present,ship_count
2. 5001,1,3
3. 5002,0,0

4. 5003,1,1

Requirements:

1. The submission file must contain exactly 3 columns: `image_id`, `ship_present`, and `ship_count`.
2. `ship_present` must be a binary value (0 or 1), and `ship_count` must be an integer greater than or equal to 0.
3. The submission must contain 1,250 rows, matching the test set, with no missing values (NaN), and must be saved as `teamname_PS4.csv`.

Evaluation Metric

Your submission will be scored on:

Metric	Description	Direction	Weight
F1-Score	Ship presence classification performance	Higher is better	60%
MAE	Ship count estimation error	Lower is better	40%

Final Score

Since F1-Score and MAE have opposite optimization directions, the combined score should be defined with a normalized or appropriately scaled MAE:

$$\text{Final Score} = (0.6 \times \text{F1-Score}) - (0.4 \times \text{Normalized MAE})$$

Higher final score wins.